PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-agent Simulation Approach with High-dimension State Space of Political Opinions

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Abstract

This pilot study examines whether inter-individual differences moderate susceptibility to cyber-manipulation in a high-dimensional political opinion state space. We implemented an ontology-driven multi-agent simulation pipeline linking PROFILE, ATTACK, and OPINION ontologies, generated baseline and post-exposure opinions with structured large-language-model agents, and estimated moderation using a parsimonious structural equation model supplemented by robust regression and bootstrap intervals. In run run_3, the pilot included 20 scenarios, balanced attack and control conditions, and one common misinformation vector. Attack-present scenarios showed a mean delta of 6.5 versus 0.0 for controls, while the primary susceptibility interaction was estimated at -21.71 ($p = 0.045$). The pipeline produced auditable realism and coherence checks, interactive exploration outputs, and print-ready publication assets. Findings are exploratory and intended to validate the methodology rather than support causal claims.

1 Introduction

Political persuasion in digitally mediated environments increasingly operates through repeated exposure, manipulative framing, and socially embedded misinformation rather than through overtly technical attacks (Vosoughi et al., 2018; Lewandowsky et al., 2017). The present project frames this problem in terms of *cognitive sovereignty*: the degree to which an individual's political opinion state remains self-governed under adversarial informational pressure. Rather than relying on inaccessible or poorly ontology-mapped empirical datasets, this pilot uses a multi-agent simulation architecture to test whether profile-level differences systematically moderate attack effectivity.

The methodological contribution is twofold. First, the pipeline represents persons, attack vectors, and political opinion items using explicit hierarchical ontologies. Second, the simulation formalizes attack effectivity as a within-scenario shift between baseline and post-exposure opinions, enabling moderation analysis at the level of individual profile configuration. This pilot therefore focuses on methodological credibility, internal coherence, and reproducibility rather than on publication-grade statistical power.

2 Materials and Methods

2.1 Study Design

Run run_3 used the test ontology triplet and generated 20 scenarios with a 50/50 control-treatment split. The treatment arm used a single common misinformation vector, `ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing`, to reduce design variance while the pipeline architecture was still being validated. Scenario creation was stratified over a derived susceptibility index so that low- and high-susceptibility profiles were represented across both conditions.

PROFILE configurations included mixed variable types, combining categorical attributes such as sex with continuous percentile-based traits derived from Big Five facets (Costa and McCrae, 1992). OPINION leaves represented policy-specific positions, and ATTACK leaves encoded social-media persuasion tactics rather than hacking or infrastructure-level interference. Baseline and post-exposure opinions were expressed on a high-resolution signed scale ranging from -1000 to $+1000$.

2.2 Agentic Pipeline

The pipeline proceeded in sequential stages: scenario creation, baseline opinion assessment, attack generation, post-attack opinion assessment, delta construction, moderation modeling, interactive visualization, publication-asset generation, and manuscript build. Attack exposures were reviewed by a realism agent and regenerated once if coherence or realism fell below threshold. Opinion assessments underwent a second self-supervision pass to penalize coarse scoring and implausible reversals.

2.3 Modeling Strategy

The primary moderator was fixed in advance as `profile_cont_susceptibility_index`. The main structural equation model followed a parsimonious pilot specification inspired by standard SEM practice (Rosseel, 2012):

```
delta_score ~ attack_present + baseline_score + susceptibility + attack_present:suscept
post_score ~ baseline_score + attack_present + susceptibility + attack_present:suscepti
```

Because pilot samples are unstable for inference, robust OLS estimates and non-parametric bootstrap intervals (Efron and Tibshirani, 1993) were reported alongside the SEM.

3 Results

Figure 1 summarizes the workflow, Figure 2 shows the treated-control delta distribution, Figure 3 visualizes the primary moderation trend, and Figure 4 summarizes primary and exploratory interaction coefficients.

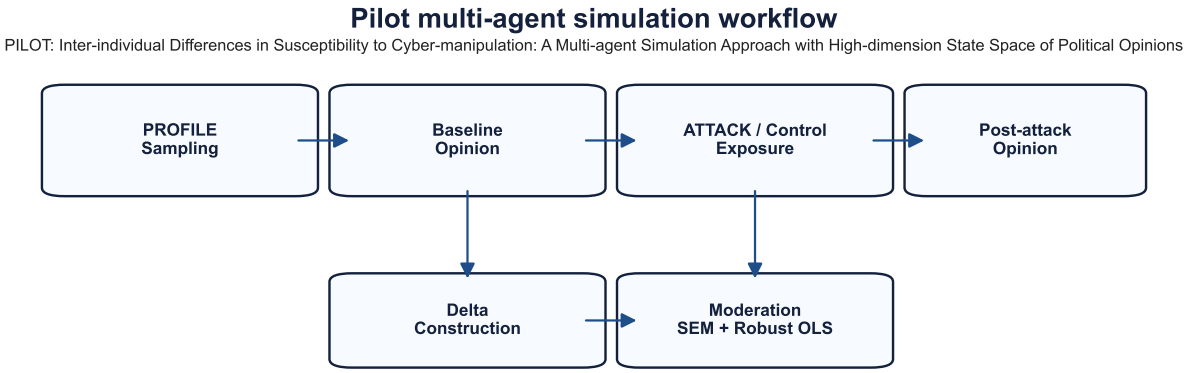

Figure 1: Figure 1. Pilot study design and pipeline schematic.

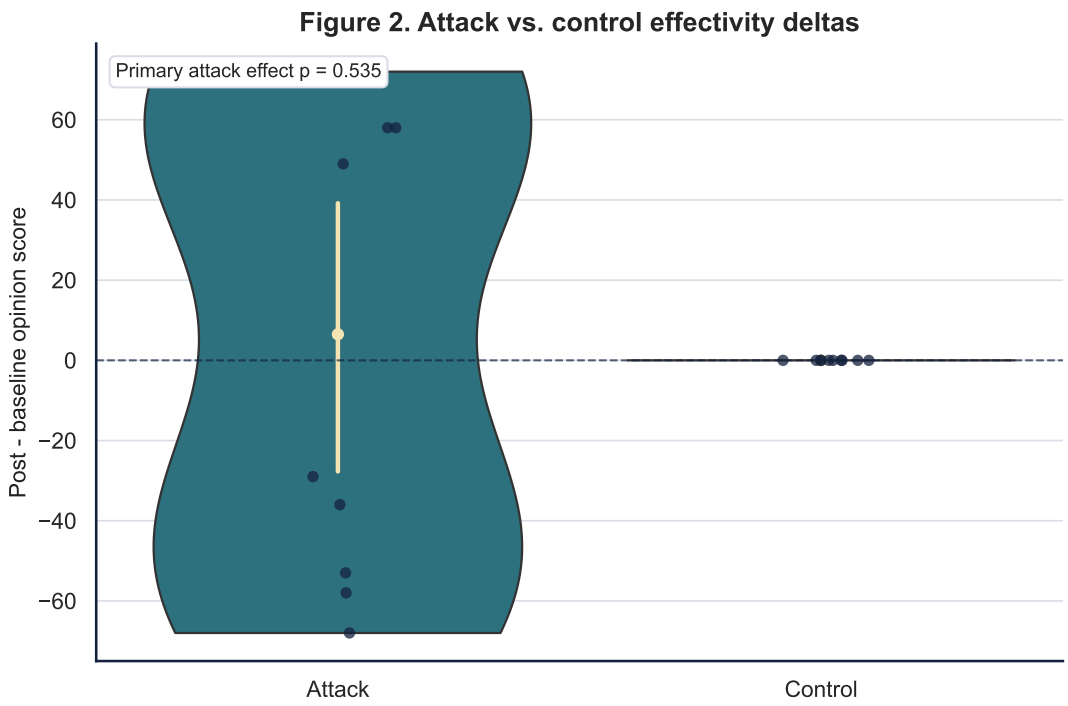

Figure 2: Figure 2. Attack versus control delta distribution with raw scenario points.

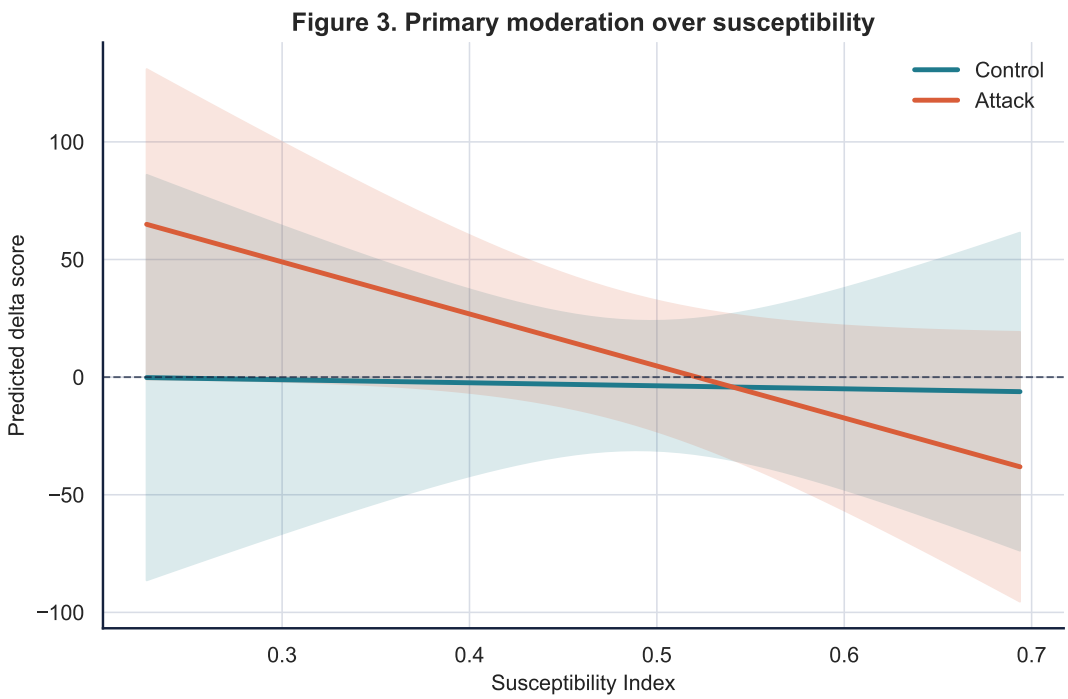

Figure 3: Figure 3. Primary moderation interaction over susceptibility.

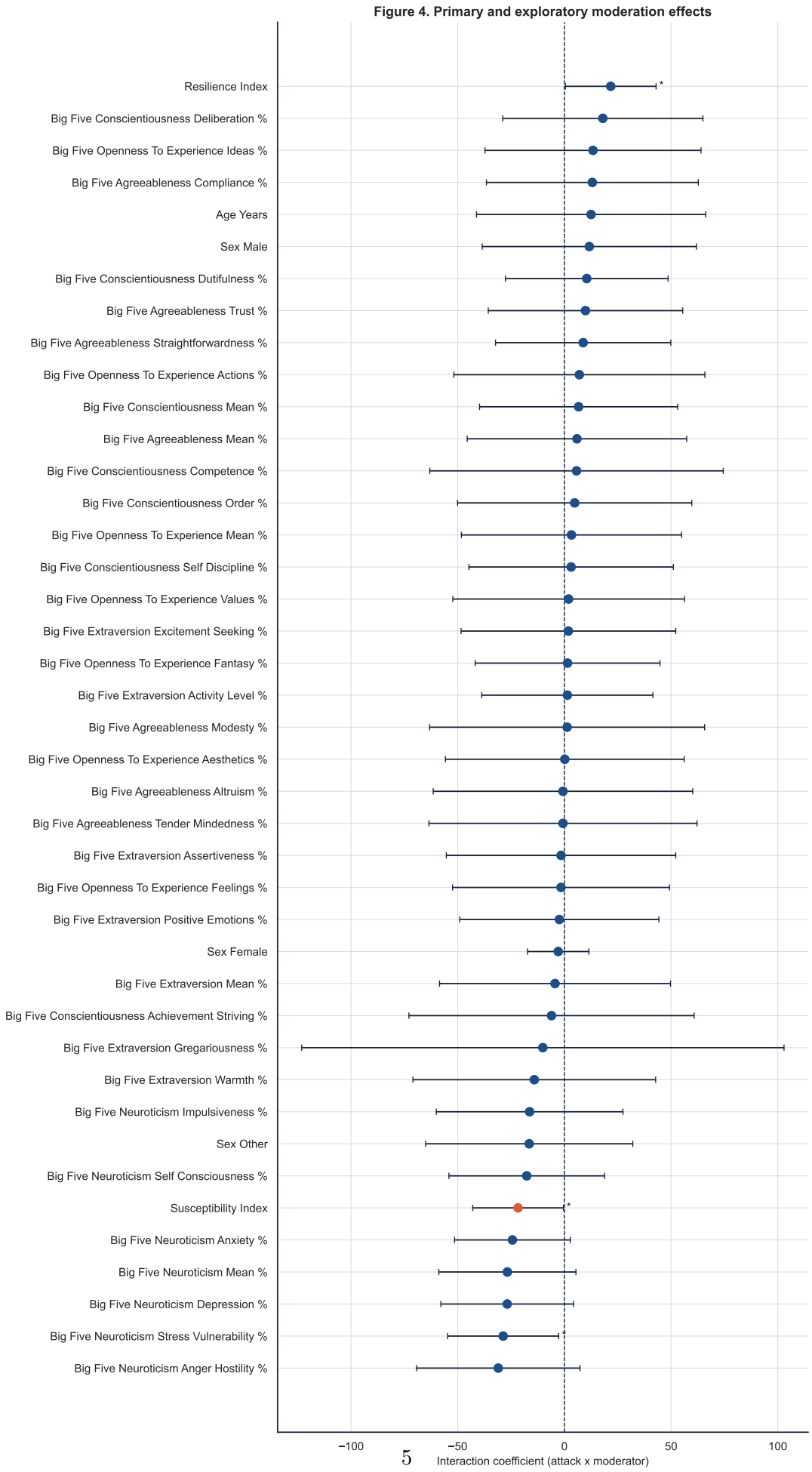

Figure 4: Figure 4. Primary and exploratory moderator interaction coefficients.

Table 1: Table 1. Pilot design and run con

Parameter	Value
Run identifier	run_3
Paper title	PILOT: Inter-individual Differences in Susceptibility to Cyber-manipulation: A Multi-
Scenarios	20
Attack ratio	0.500
Attack leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Fra
OpenRouter model	mistralai/mistral-small-3.2-24b-instruct
Primary moderator	profile_cont_susceptibility_index
Bootstrap samples	500
Self-supervised attack realism	True
Test ontology	True

Note. Run-level configuration for the pilot simulation reported in run 3.

Table 2: Table 2. Descriptive statistics by condition

Condition	n	baseline_mean	post_mean	delta_mean	delta_sd	susceptibility_mean	realism_mean
Attack	10	365.900	372.400	6.500	59.636	0.476	0.780
Control	10	236.000	236.000	0.000	0.000	0.494	0.000

Note. Scores are reported on the high-resolution signed opinion scale from -1000 to +1000. Realism scores apply only to treated scenarios.

Table 3: Table 3. Primary moderation model with robust and boo

term	estimate	std_error	p_value	conf_low_x	conf_high_x	bootstrap_mean	b
Intercept	13.233	12.650	0.296	-11.561	38.027	14.305	
attack_present	11.558	18.619	0.535	-24.934	48.050	13.171	
baseline_score	-0.056	0.038	0.140	-0.129	0.018	-0.064	
primary_moderator_z	-1.338	7.024	0.849	-15.105	12.430	-3.156	
attack_x_primary_moderator	-21.707	10.848	0.045	-42.968	-0.446	-21.069	

Note. Robust standard errors are HC3. Bootstrap intervals are percentile intervals from resampled pilot fits and are exploratory

The attack condition shifted opinions more strongly than the control condition in directional terms (mean treated delta = 6.5; mean control delta = 0.0). The robust attack-present coefficient was 11.56 with p = 0.535, while the primary moderation coefficient was -21.71 with p = 0.045. Baseline opinion was comparatively stable across models (robust estimate = -0.056, p = 0.140).

The SEM converged, but fit remained pilot-limited (CFI = 0.080, RMSEA = 1.817). Treated exposures achieved a mean realism score of 0.78, which indicates that the self-supervised reviewer generally accepted the generated persuasion content as platform-plausible. The most informative exploratory moderators were Big Five Neuroticism Stress Vulnerability

4 Discussion

The present pilot demonstrates that the pipeline works end to end: ontology-guided scenario generation, structured LLM assessments, realism/coherence review loops, moderation estimation, static figure export, interactive SEM inspection, and automated manuscript generation all completed within a single reproducible workflow. This is the main result of the pilot.

At the same time, the inferential limitations are substantial. With only 20 scenarios, the moderation coefficients are far too unstable for substantive claims about real-world susceptibility. The objective of this pilot is therefore methodological validation, not empirical estimation. The directional separation between attack and control conditions is encouraging because it suggests that the attack-generation and opinion-assessment components are not degenerate. However, the uncertainty around the interaction term means that profile moderation remains a hypothesis to be tested at scale rather than a result established here.

Three methodological design decisions improved the credibility of the pilot. First, the primary moderator was fixed in advance rather than selected opportunistically from whichever variable produced the strongest coefficient. Second, attack realism and opinion coherence were explicitly audited with repair loops. Third, the report package preserves provenance, stage manifests, and static/interactice outputs together, making the pipeline easier to scrutinize or reproduce.

5 Conclusion

This pilot supports the feasibility of an ontology-driven multi-agent simulation framework for studying how inter-individual differences may moderate susceptibility to cyber-manipulation in political opinion spaces. The pipeline is operational, auditable, and sufficiently modular to scale beyond the present pilot. The next step is not interpretive escalation but design expansion: more scenarios, multiple attack leaves, and cross-model sensitivity checks.

A Supplementary Materials

The repository also includes an interactive SEM dashboard generated from run run_3. The static publication assets below are the paper-ready subset of that richer exploratory output.

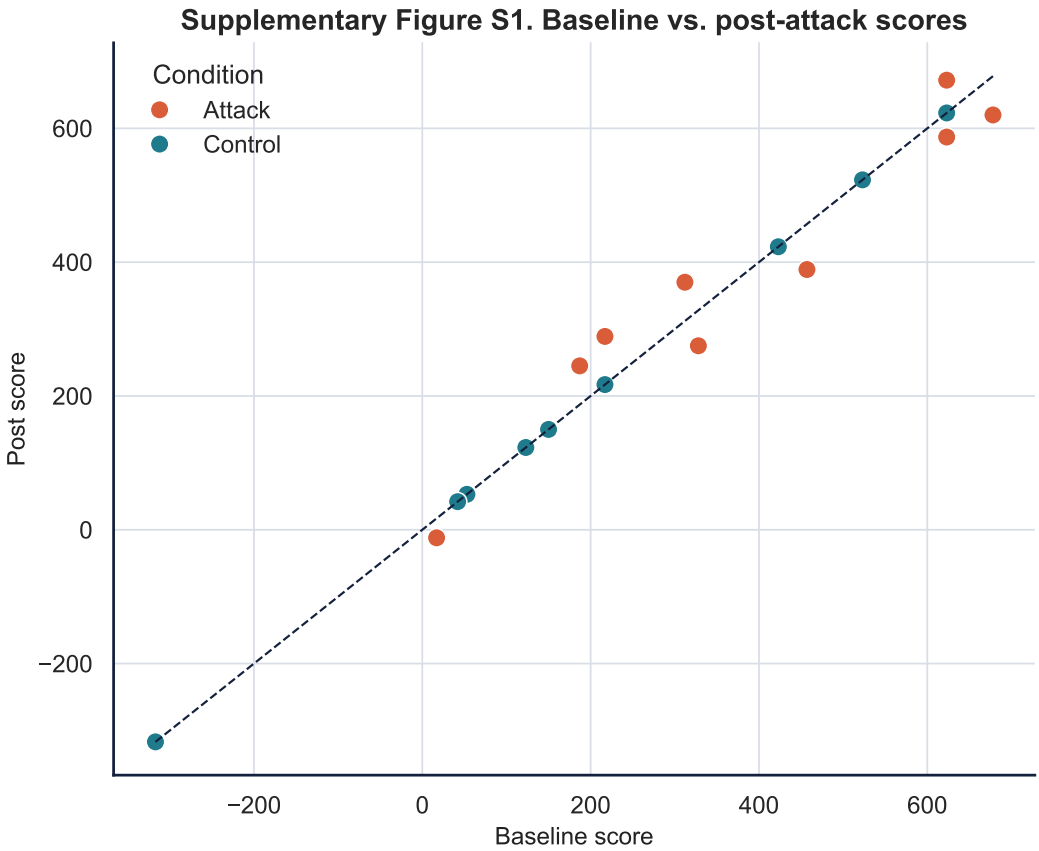

Figure 5: Supplementary Figure S1. Baseline versus post-attack opinion scores.

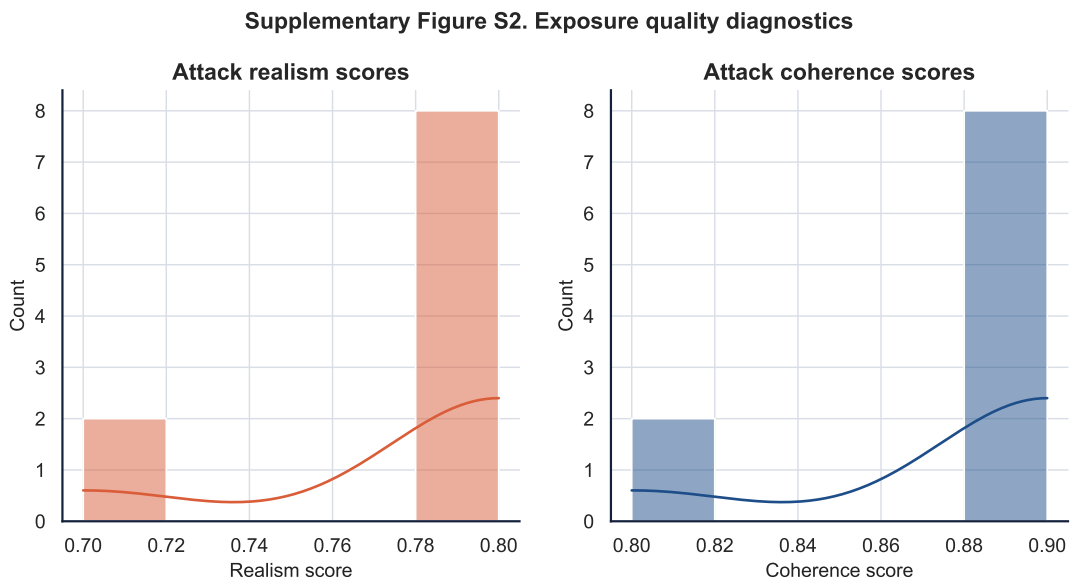

Figure 6: Supplementary Figure S2. Attack realism and coherence distributions.

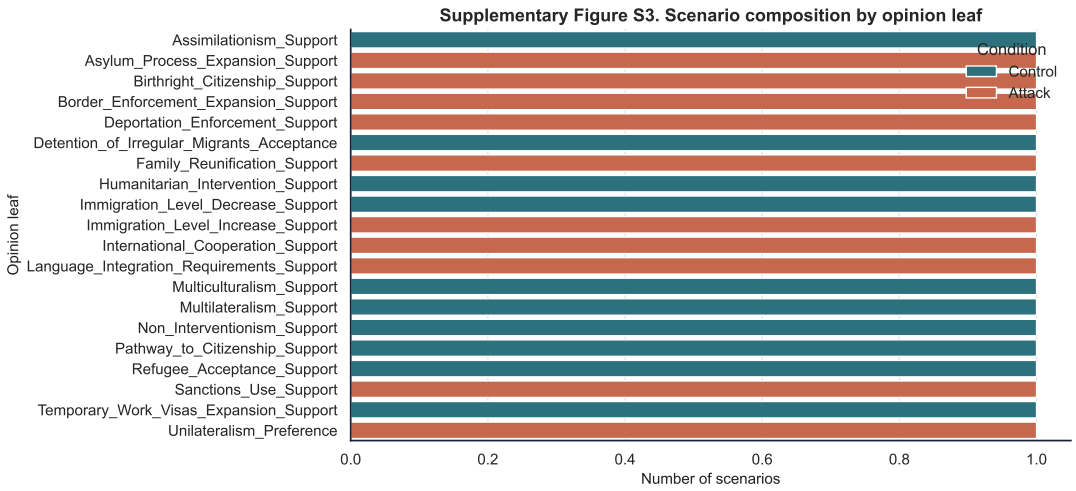

Figure 7: Supplementary Figure S3. Scenario composition across opinion leaves.

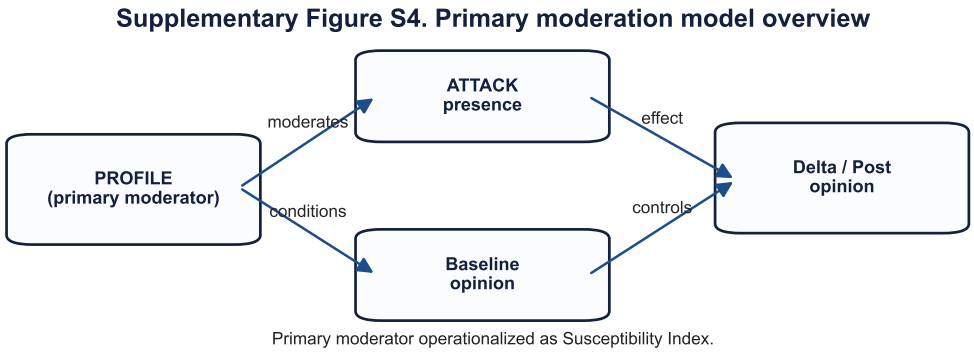

Figure 8: Supplementary Figure S4. Primary moderation model overview.

References

Costa, P. T. and McCrae, R. R. (1992). *Neo PI-R Professional Manual*. Psychological Assessment Resources.

Efron, B. and Tibshirani, R. J. (1993). *An Introduction to the Bootstrap*. Chapman & Hall/CRC.

Lewandowsky, S., Ecker, U. K. H., and Cook, J. (2017). Beyond misinformation: Understanding and coping with the post-truth era. *Journal of Applied Research in Memory and Cognition*, 6(4):353–369.

Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, 48(2):1–36.

Vosoughi, S., Roy, D., and Aral, S. (2018). The spread of true and false news online. *Science*, 359(6380):1146–1151.

Table 4: Supplementary Table S1. Ontology leaves used

ontology	leaf_label	leaf_path
OPINION	Humanitarian_Intervention_Support	Issue_Position_Taxonomy > Foreign_Policy_and_Ge
OPINION	International_Cooperation_Support	Issue_Position_Taxonomy > Foreign_Policy_and_Ge
OPINION	Multilateralism_Support	Issue_Position_Taxonomy > Foreign_Policy_and_Ge
OPINION	Non_Interventionism_Support	Issue_Position_Taxonomy > Foreign_Policy_and_Ge
OPINION	Sanctions_Use_Support	Issue_Position_Taxonomy > Foreign_Policy_and_Ge
OPINION	Unilateralism_Preference	Issue_Position_Taxonomy > Foreign_Policy_and_Ge
OPINION	Assimilationism_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Asylum_Process_Expansion_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Birthright_Citizenship_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Border_Enforcement_Expansion_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Deportation_Enforcement_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Detention_of_Irregular_Migrants_Acceptance	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Family_Reunification_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Immigration_Level_Decrease_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Immigration_Level_Increase_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Language_Integration_Requirements_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Multiculturalism_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Pathway_to_Citizenship_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Refugee_Acceptance_Support	Issue_Position_Taxonomy > Immigration_and_Citize
OPINION	Temporary_Work_Visas_Expansion_Support	Issue_Position_Taxonomy > Immigration_and_Citize
ATTACK	Misleading narrative framing	ATTACK_VECTORS > Social_Media_Misinformatio

Note. Opinion leaves reflect the scenario allocation in run 3; the attack ontology was restricted to one common misinformation v

moderator_column	moderator_label
profile_cont_big_five_neuroticism_stress_vulnerability_pct	Big Five Neuroticism Stress Vulnerability %
profile_cont_resilience_index	Resilience Index
profile_cont_big_five_neuroticism_anxiety_pct	Big Five Neuroticism Anxiety %
profile_cont_big_five_neuroticism_depression_pct	Big Five Neuroticism Depression %
profile_cont_big_five_neuroticism_mean_pct	Big Five Neuroticism Mean %
profile_cont_big_five_neuroticism_anger_hostility_pct	Big Five Neuroticism Anger Hostility %
profile_cont_big_five_neuroticism_self_consciousness_pct	Big Five Neuroticism Self Consciousness %
profile_cont_big_five_conscientiousness_deliberation_pct	Big Five Conscientiousness Deliberation %
profile_cont_big_five_neuroticism_impulsiveness_pct	Big Five Neuroticism Impulsiveness %
profile_cat___profile_cat_sex_Other	Sex Other
profile_cont_big_five_conscientiousness_dutifulness_pct	Big Five Conscientiousness Dutifulness %
profile_cont_big_five_openness_to_experience_ideas_pct	Big Five Openness To Experience Ideas %
profile_cont_big_five_agreeableness_compliance_pct	Big Five Agreeableness Compliance %
profile_cont_big_five_extraversion_warmth_pct	Big Five Extraversion Warmth %
profile_cont_age_years	Age Years
profile_cat___profile_cat_sex_Male	Sex Male
profile_cont_big_five_agreeableness_trust_pct	Big Five Agreeableness Trust %
profile_cont_big_five_agreeableness_straightforwardness_pct	Big Five Agreeableness Straightforwardness %
profile_cat___profile_cat_sex_Female	Sex Female
profile_cont_big_five_conscientiousness_mean_pct	Big Five Conscientiousness Mean %
profile_cont_big_five_openness_to_experience_actions_pct	Big Five Openness To Experience Actions %
profile_cont_big_five_agreeableness_mean_pct	Big Five Agreeableness Mean %
profile_cont_big_five_conscientiousness_achievement_striving_pct	Big Five Conscientiousness Achievement Strivi
profile_cont_big_five_extraversion_gregariousness_pct	Big Five Extraversion Gregariousness %
profile_cont_big_five_conscientiousness_order_pct	Big Five Conscientiousness Order %
profile_cont_big_five_conscientiousness_competence_pct	Big Five Conscientiousness Competence %
profile_cont_big_five_extraversion_mean_pct	Big Five Extraversion Mean %
profile_cont_big_five_conscientiousness_self_discipline_pct	Big Five Conscientiousness Self Discipline %
profile_cont_big_five_openness_to_experience_mean_pct	Big Five Openness To Experience Mean %
profile_cont_big_five_extraversion_positive_emotions_pct	Big Five Extraversion Positive Emotions %
profile_cont_big_five_extraversion_excitement_seeking_pct	Big Five Extraversion Excitement Seeking %
profile_cont_big_five_openness_to_experience_values_pct	Big Five Openness To Experience Values %
profile_cont_big_five_extraversion_activity_level_pct	Big Five Extraversion Activity Level %
profile_cont_big_five_openness_to_experience_fantasy_pct	Big Five Openness To Experience Fantasy %
profile_cont_big_five_openness_to_experience_feelings_pct	Big Five Openness To Experience Feelings %
profile_cont_big_five_extraversion_assertiveness_pct	Big Five Extraversion Assertiveness %
profile_cont_big_five_agreeableness_modesty_pct	Big Five Agreeableness Modesty %
profile_cont_big_five_agreeableness_altruism_pct	Big Five Agreeableness Altruism %
profile_cont_big_five_agreeableness_tender_mindedness_pct	Big Five Agreeableness Tender Mindedness %
profile_cont_big_five_openness_to_experience_aesthetics_pct	Big Five Openness To Experience Aesthetics %
profile_cont_susceptibility_index	Susceptibility Index

Note. Each moderator was estimated in a separate attack-by-moderator interaction model with baseline opinion included as a co

Table 6: Supplementary Table S3. Assumption an

section	item	status
Assumption register	Pilot sample size is sufficient for stable moderation estimates.	risk
Assumption register	Treatment-control balance is adequate.	ok
Assumption register	Opinion measurement has enough numeric resolution.	ok
Assumption register	Attack text realism is adequate.	ok
Assumption register	SEM converges with acceptable fit.	ok
Assumption register	LLM outputs are reproducible and auditable.	ok
Peer-review critique	Synthetic LLM agents may not represent human cognition or causal response behavior.	addres
Peer-review critique	Adversarial content may be unrealistic or too generic.	addres
Peer-review critique	Model dependence on a single LLM could bias outputs.	addres
Peer-review critique	Insufficient sample size inflates uncertainty and may destabilize SEM.	addres

Note. This table summarizes the principal methodological assumptions and the explicit mitigations implemented in the pilot pip

Table 7: Supplementary Table S4. Reproducibility manifest

Field	Value
run_id	run_3
output_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/evaluati
ontology_root	/Users/stijnvanseveren/PythonProjects/Paper_CaseStudiesAnalysisExperimentalData/src/back
attack_leaf	ATTACK_VECTORS > Social_Media_Misinformation > Misleading_Narrative_Framing
openrouter_model	mistralai/mistral-small-3.2-24b-instruct
primary_moderator	profile_cont_susceptibility_index
opinion_leaf_count	38
attack_leaf_count	9

Note. Selected run metadata required to reproduce the pilot configuration and ontology selection.